

# ALI RAZA

## Materials and Metallurgy Engineer



### About Me

Versatile and results-driven Metallurgy and Materials Engineer with extensive expertise in materials processing, advanced welding techniques such as Friction Stir Welding (FSW), TIG, MIG, and Arc welding, as well as comprehensive metallurgical analysis. Proficient in melting, casting, and optimizing metal alloys to meet stringent quality standards. Renowned for delivering innovative solutions, enhancing material performance, and driving efficiency in demanding industrial operations. Ready to make a significant impact in high-performance engineering roles.



+966576324172



enr.aliraza272@gmail.com



Riyadh, Saudi Arabia

### EXPERTISE

- Materials Testing and Characterization
- Metals/Alloys Casting and Forging
- Metallurgical Analysis
- Advanced welding techniques: Friction Stir Welding (FSW), TIG, MIG, and Arc Welding
- Materials processing, melting, and casting of steel, aluminum, and other alloys
- Corrosion Control and Protection
- Foundry Engineering
- Surface Engineering
- Production and Operational Management
- AutoCAD, Microsoft Office, Matlab, Aynsys

### Soft Skills

- Strategic Problem Solving
- Critical Thinking & Decision
- Effective Communication
- Leadership & Team
- Adaptability & Resilience
- Proactive & Initiative-Driven

### EDUCATION

**Institute Of Metallurgy and Materials Engineering University of the Punjab Lahore , Pakistan**

Bachelor's of Metallurgy and Materials Engineering

### Certifications

**Certificate Of Registered Engineer In Pakistan**

**Engineering Council (PEC) :Registered**

**Professional Engineer(PE) : Licence**

**Course Study of Health and Safety at**

**IOSH Acedmy.**

### LANGUAGE

- English (Fluent)

### EXPERIENCE

#### Hadeed Pakistan Pvt.Ltd (Trainee Engineer) (2023-2024)

- Material Testing and Quality Control
- Failure Analysis and Troubleshooting
- Failure and Corrosion Analysis
- Heat treatment Batch Annealing furnace (BAF)
- Cold Rolling Mill (6i)
- Ensuring that the materials meet stringent quality standards.
- I work closely with the engineering team to optimize production efficiency
- Monitor metallurgical processes
- Enhancing the overall output and quality of steel products
- Reduce waste, and improve the overall quality of steel products

#### Final Year Project (FYP)

##### Welding Processes

- FSW,TIG,MIG,SMAW,SAW,GTAW

##### Enhance FSW Weld Quality

- Friction Stir welding (FSW) at Aluminum 6061 enhance weld quality at Aluminum 6061 by using Different welding tools design (Plain pin configuration and Threatened pin configuration)

##### Comparison between Traditional & Advance Welding techniques

- Advance Welding techniques with traditional Welding techniques.
- Mechanical Testing Techniques (Hardness, Tensile test, 3-Point Band test.
- NDT Techniques (UT, VT, MPT, RT, DPT)
- Characterization Techniques (XRD, XRF, SEM, TEM, OM)

### KNOWLEDGE OF STEEL GRADE AND SPECIFICATIONS

1. Regulatory Knowledge (ASTM International Standards)
2. ISO (International Organization for Standardization) Standards (e.g., ISO 4948, ISO 683, ISO 9001)
3. AISI (American Iron and Steel Institute)
4. SAE (Society of Automotive Engineers) Standards

### ADDITIONAL INFORMATION

Saudi Arabia Driving liscence: Valid

Iqama : Transferable

Open to work remotely all over the Kingdom Of Saudi Arabia

# ALI RAZA

## Materials and Metallurgy Engineer

+966-57-6324172  
Engr.aliraza272@gmail.com  
Riyadh, Saudi Arabia

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Hiring Manager, Really Great Place

Dear HR,

I am writing to express my strong interest in the Engineering position that recently opened at your Company as a Materials Engineer. As a highly motivated and results-oriented Materials and Metallurgy Engineer with a proven track record of success in the field, I am confident that I possess the skills and experience you are looking for.

My expertise lies in materials processing, advanced welding techniques, and metallurgical analysis. My experience at Hadeed Pakistan Pvt. Ltd. has equipped me with a strong foundation in these areas.

- Mechanical testing and Materials inspection
- Welding Techniques
- Melting and Casting of Metals and Alloys
- Heat Treatment processes (Quenching, Annealing, Tempering, Normalizing, etc)
- Annealing furnace (BAF)
- Rolling Mills
- Corrosion protection

I am proficient in Mechanical testing, Materials inspection, AutoCAD, and Microsoft Office. I have a deep understanding of the Manufacturing and Construction Steel Structure Industries. I am eager to contribute my skills and knowledge to a challenging and rewarding role where I can continue to grow and develop my expertise.

Thank you for your time and consideration. I have attached my updated resume for your review.

Sincerely,

Ali Raza

*Ali Raza*



**UNIVERSITY OF THE PUNJAB**  
Faculty of Chemical and Materials Engineering  
**Institute of Metallurgy & Materials Engineering**  
B.Sc. (Engg.) Metallurgy and Materials Engineering



**Academic Transcript of B.Sc. (Engg.) Metallurgy and Materials Engineering**

Student Name: **ALI RAZA**  
Father's Name: **MUHAMMAD SHAFIQ**

Roll No.: **20MME-S2-311**  
Reg. No.: **2020-UEE-56**  
Result: **Pass**

Session: **Batch 2020-2024**  
Notification No.: **1** Date: **01-10-2024**  
Status: **Degree Completed**

| Course Code                                                 | Course Title                                         | CH      | G.P  | GRD | Total | Obtd. | Obt % |
|-------------------------------------------------------------|------------------------------------------------------|---------|------|-----|-------|-------|-------|
| <b>First Semester - Fall 2020</b>                           |                                                      |         |      |     |       |       |       |
| HUM 111                                                     | Functional English                                   | 3       | 3.30 | B+  | 300   | 231   | 77    |
| NSC 112                                                     | Applied Chemistry                                    | 3       | 2.30 | C+  | 300   | 189   | 63    |
| NSC 112L                                                    | Applied Chemistry (Lab)                              | 1       | 3.70 | A-  | 100   | 82    | 82    |
| NSC 113                                                     | Engineering Mathematics-I                            | 3       | 2.70 | B-  | 300   | 195   | 65    |
| CSC 114                                                     | Computer Science and Information Technology          | 2       | 3.00 | B   | 200   | 148   | 74    |
| CSC 114L                                                    | Computer Science and Information Technology (Lab)    | 1       | 4.00 | A   | 100   | 93    | 93    |
| MME 115                                                     | Fundamentals of Metallurgy and Materials Engineering | 3       | 1.00 | D   | 300   | 153   | 51    |
| HUM 116                                                     | Islamic Studies                                      | 2       | 3.00 | B   | 200   | 144   | 72    |
| CH: 18 CCH: 18 Marks: 1235 PM: 68.61% GPA: 2.64 CGPA: 2.64  |                                                      |         |      |     |       |       |       |
| <b>Second Semester - Spring 2021</b>                        |                                                      |         |      |     |       |       |       |
| MME 121L                                                    | Engineering Drawing                                  | 1       | 4.00 | A   | 100   | 86    | 86    |
| MME 122L                                                    | Workshop Practice                                    | 1       | 3.00 | B   | 100   | 71    | 71    |
| MIN 123                                                     | Mineral Processing                                   | 2       | 2.70 | B-  | 200   | 130   | 65    |
| MIN 123L                                                    | Mineral Processing (Lab)                             | 1       | 3.70 | A-  | 100   | 80    | 80    |
| NSC 124N                                                    | Engineering Mathematics-II                           | 3       | 2.00 | C   | 300   | 174   | 58    |
| NSC 125 *                                                   | Applied Physics                                      | 3       | 3.00 | B   | 300   | 210   | 70    |
| NSC 125L                                                    | Applied Physics (Lab)                                | 1       | 3.30 | B+  | 100   | 76    | 76    |
| HUM 126                                                     | Pakistan Studies                                     | 2       | 2.70 | B-  | 200   | 132   | 66    |
| HUM 127                                                     | The Holy Quran Translation                           | 1       | 4.00 | A   | 100   | 90    | 90    |
| CH: 15 CCH: 33 Marks: 1049 PM: 69.93% GPA: 2.92 CGPA: 2.77  |                                                      |         |      |     |       |       |       |
| <b>Third Semester - Fall 2021</b>                           |                                                      |         |      |     |       |       |       |
| HUM 211                                                     | Communication skills                                 | 2       | 3.70 | A-  | 200   | 160   | 80    |
| MME 212 *                                                   | Physical Metallurgy                                  | 3       | 3.00 | B   | 300   | 210   | 70    |
| MME 212L                                                    | Physical Metallurgy (Lab)                            | 1       | 3.00 | B   | 100   | 72    | 72    |
| MME 213 *                                                   | Ceramics Science and Engineering                     | 3       | 1.00 | D   | 300   | 153   | 51    |
| MME 213L                                                    | Ceramics Science and Engineering (Lab)               | 1       | 3.00 | B   | 100   | 70    | 70    |
| MME 214N *                                                  | Engineering Mechanics                                | 3       | 3.00 | B   | 300   | 210   | 70    |
| MSC 215                                                     | Industrial Safety and Environmental Management       | 3       | 3.00 | B   | 300   | 210   | 70    |
| CH: 16 CCH: 49 Marks: 1085 PM: 67.81% GPA: 2.71 CGPA: 2.75  |                                                      |         |      |     |       |       |       |
| <b>Fourth Semester - Spring 2022</b>                        |                                                      |         |      |     |       |       |       |
| MME 221                                                     | Mechanical Behavior of Engineering Materials         | 3       | 1.70 | C-  | 300   | 165   | 55    |
| MME 221L                                                    | Mechanical Behavior of Engineering Materials (Lab)   | 1       | 3.30 | B+  | 100   | 75    | 75    |
| MME 222N                                                    | Polymer Science and Engineering                      | 3       | 1.00 | D   | 300   | 153   | 51    |
| MME 223N *                                                  | Materials Thermodynamics and Kinetics                | 3       | 2.70 | B-  | 300   | 195   | 65    |
| MME 224N *                                                  | Foundry Engineering-I                                | 2       | 1.00 | D   | 200   | 102   | 51    |
| MME 224L                                                    | Foundry Engineering-I (Lab)                          | 1       | 3.00 | B   | 100   | 72    | 72    |
| CSC 225L                                                    | Computer Aided Design (Lab)                          | 1       | 1.00 | D   | 100   | 53    | 53    |
| CSC 226L                                                    | MATLAB and SIMULINK (Lab)                            | 1       | 2.30 | C+  | 100   | 62    | 62    |
| HUM 227                                                     | Technical Writing                                    | 2       | 1.00 | D   | 200   | 102   | 51    |
| HUM 228                                                     | The Holy Quran Translation                           | 1       | 4.00 | A   | 100   | 91    | 91    |
| CH: 18 CCH: 67 Marks: 1070 PM: 59.44% GPA: 1.88 CGPA: 2.52  |                                                      |         |      |     |       |       |       |
| <b>Fifth Semester - Fall 2022</b>                           |                                                      |         |      |     |       |       |       |
| MME 311N                                                    | Iron Manufacturing Technology                        | 2       | 4.00 | A   | 200   | 170   | 85    |
| MME 312N                                                    | Foundry Engineering-II                               | 3       | 1.00 | D   | 300   | 153   | 51    |
| MME 312N L                                                  | Foundry Engineering-II Lab                           | 1       | 4.00 | A   | 100   | 89    | 89    |
| MME 313N                                                    | Heat Treatment of Metals and Alloys                  | 3       | 2.30 | C+  | 300   | 189   | 63    |
| MME 313L N                                                  | Heat Treatment of Metals and Alloys (Lab)            | 1       | 3.70 | A-  | 100   | 80    | 80    |
| MME 314N *                                                  | Transport Processes                                  | 3       | 2.70 | B-  | 300   | 195   | 65    |
| MME 315N                                                    | Metallurgical Manufacturing Processes                | 3       | 2.70 | B-  | 300   | 195   | 65    |
| HUM 316                                                     | Critical Thinking and Reasoning                      | 2       | 3.30 | B+  | 200   | 150   | 75    |
| CH: 18 CCH: 85 Marks: 1221 PM: 67.83% GPA: 2.69 CGPA: 2.55  |                                                      |         |      |     |       |       |       |
| <b>Sixth Semester - Spring 2023</b>                         |                                                      |         |      |     |       |       |       |
| MME 321                                                     | Corrosion Engineering                                | 3       | 1.00 | D   | 300   | 150   | 50    |
| MME 321L                                                    | Corrosion Engineering (Lab)                          | 1       | 3.70 | A-  | 100   | 80    | 80    |
| MME 322                                                     | Welding and Joining Processes                        | 3       | 3.00 | B   | 300   | 213   | 71    |
| MME 322L                                                    | Welding and Joining Processes (Lab)                  | 1       | 2.70 | B-  | 100   | 66    | 66    |
| MME 323N                                                    | Steel Manufacturing Processes                        | 3       | 3.70 | A-  | 300   | 240   | 80    |
| MME 324N                                                    | Composite Materials                                  | 2       | 3.70 | A-  | 200   | 160   | 80    |
| MME 325L                                                    | Polymeric and Composite Materials (Lab)              | 1       | 2.30 | C+  | 100   | 63    | 63    |
| CSC 326                                                     | Computational Materials Science                      | 2       | 1.00 | D   | 200   | 106   | 53    |
| CSC 327L                                                    | Computer Applications in Materials Engineering (Lab) | 1       | 2.30 | C+  | 100   | 62    | 62    |
| HUM 328                                                     | The Holy Quran Translation                           | 1       | 4.00 | A   | 100   | 96    | 96    |
| CH: 18 CCH: 103 Marks: 1236 PM: 68.67% GPA: 2.64 CGPA: 2.57 |                                                      |         |      |     |       |       |       |
| <b>Seventh Semester - Fall 2023</b>                         |                                                      |         |      |     |       |       |       |
| MME 411                                                     | Tribology and Surface Engineering                    | 3       | 3.00 | B   | 300   | 210   | 70    |
| MME 411L                                                    | Tribology and Surface Engineering (Lab)              | 1       | 3.70 | A-  | 100   | 83    | 83    |
| MME 412N                                                    | Materials Characterization                           | 2       | 1.70 | C-  | 200   | 110   | 55    |
| MME 412NL                                                   | Materials Characterization (Lab)                     | 1       | 4.00 | A   | 100   | 88    | 88    |
| MME 413N                                                    | Non-Ferrous Metallurgy                               | 3       | 2.70 | B-  | 300   | 204   | 68    |
| MME 414B                                                    | Nanomaterials and Nanotechnology                     | 2       | 3.00 | B   | 200   | 140   | 70    |
| MSC 415N                                                    | Industrial Quality Management                        | 3       | 2.30 | C+  | 300   | 186   | 62    |
| MME 416N                                                    | Design Project-I                                     | 3       | 4.00 | A   | 300   | 258   | 86    |
| CH: 18 CCH: 121 Marks: 1279 PM: 71.06% GPA: 2.95 CGPA: 2.62 |                                                      |         |      |     |       |       |       |
| <b>Eighth Semester - Spring 2024</b>                        |                                                      |         |      |     |       |       |       |
| MME 421                                                     | Powder Metallurgy                                    | 2       | 3.30 | B+  | 200   | 152   | 76    |
| MME 422A                                                    | Advanced Metallic Materials                          | 2       | 2.70 | B-  | 200   | 130   | 65    |
| MSC 423N                                                    | Production and Operations Management                 | 3       | 2.30 | C+  | 300   | 186   | 62    |
| MSC 424                                                     | Entrepreneurship                                     | 2       | 2.30 | C+  | 200   | 124   | 62    |
| HUM 425                                                     | Industrial Psychology and Sociology                  | 2       | 2.00 | C   | 200   | 116   | 58    |
| MME 426                                                     | Design Project - II                                  | 3       | 4.00 | A   | 300   | 258   | 86    |
| HUM 427                                                     | The Holy Quran Translation                           | 1       | 4.00 | A   | 100   | 92    | 92    |
| CH: 15 CCH: 136 Marks: 1058 PM: 70.53% GPA: 2.90 CGPA: 2.66 |                                                      |         |      |     |       |       |       |
| Total Credit Hours                                          |                                                      | 136     |      |     |       |       |       |
| Cumulative Grade Point Average (CGPA)                       |                                                      | 2.66    |      |     |       |       |       |
| Overall Percentage Marks (OPM)                              |                                                      | 67.89 % |      |     |       |       |       |
| Obtained Marks                                              |                                                      | 9233    |      |     |       |       |       |
| Total Marks                                                 |                                                      | 13600   |      |     |       |       |       |

\*\*\*\*\* END OF TRANSCRIPT \*\*\*\*\*

Date/Time of Issuance: 2024-10-02 12:56:33

**Attainment of PLOs**

| PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 | PLO 8 | PLO 9 | PLO 10 | PLO 11 | PLO 12 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|--------|--------|
| 65 %  | 66 %  | 73 %  | 76 %  | 68 %  | 72 %  | 75 %  | 76 %  | 78 %  | 74 %   | 77 %   | 70 %   |

\* The subject is passed in more than one attempt

The expiation of Program Learning Outcomes (PLO) and Grading System is printed overleaf.

**Examination/Program Coordinator**

**Prepared By**

**Chairperson/Director/Principal**

**Checked By**

**Controller of Examinations (PU)**

Serial No. **317748**

Form PEC-7

Registration No: **METAL/007328**

Date of Registration: **Dec 09, 2024**



## PAKISTAN ENGINEERING COUNCIL CERTIFICATE OF REGISTRATION

(UNDER PAKISTAN ENGINEERING COUNCIL ACT 1976)

### REGISTERED ENGINEER

This is to certify that **ENGR. ALI RAZA** son of **MUHAMMAD SHAFIQ** born on **OCT 09, 2001** resident of **CHAK NO:272/H.R FORT ABBAS, DISTRICT BAHAWALNAGAR** has been registered with Pakistan Engineering Council as **REGISTERED ENGINEER** on the basis of following particulars:-

Engineering Qualification: **B.Sc. METALLURGY AND MATERIALS ENGINEERING / 2024 - INSTITUTE OF METALLURGY AND MATERIALS ENGINEERING (FORMERLY COLLEGE OF ENGINEERING AND EMERGING TECHNOLOGIES, UNIVERSITY OF THE PUNJAB, LAHORE)**

Date of Issue: Dec 20, 2024



Registrar

**Note:** This registration certificate is valid until **DEC 31, 2024** unless renewed further.  
This certificate is property of Council, returnable on request.



November 02, 2023

**To Whom It May Concern**

This letter is to certify that Mr. Ali Raza Having CNIC 31103-4858550-7 has successfully completed his internship from July 21, 2023 to Aug 18, 2023 with us on our plant.

During the span, we found him punctual and hardworking person. His association with us was beneficial.

We wish him all the best in his future endeavors.

Sincerely,

Asim Latif  
Manager Admin/HR